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Driving Sustainable Production in Malaysia's Deepwater: Field-Proven Artificial Lift Solutions for Maximizing Asset Value

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Abstract

Malaysia's deepwater assets face mounting challenges as fields mature, declining reservoir pressure, complex multiphase flows, and economic constraints threaten production sustainability. Conventional artificial lift solutions are often hampered by subsea infrastructure limitations and high intervention costs. This paper aims to share insights and field-proven strategies on innovative artificial lift technologies tailored to Malaysia's deepwater environment, driving sustainable production and maximizing asset value. A strategic review of artificial lift deployments across Malaysia's deepwater fields was conducted, focusing on technologies overcoming infrastructure and economic barriers. Key solutions include Multiphase Pumps with Compact Separator, Riser-Based Gas Lift (RBGL), Subsurface Gas Lift (SSGL) and Coiled Tubing Gas Lift (CTGL). Technology selection was guided by field-specific performance, facility constraints, and cost-efficiency. Data was gathered from multiple deepwater projects, covering technology concepts, intervention records and system integration challenges. Collaborative planning between asset teams, technology providers, and host authorities ensured fit-for-purpose solutions addressing space, flow assurance, and operational complexities. The deployment of these artificial lift solutions delivered tangible benefits and collectively extended field life, increased production rates, improved recovery factors, and optimized OPEX. Case studies demonstrated deferred abandonment and enhanced economic returns, supporting PETRONAS' goal for long-term production sustainability.

Introduction

Intensification of Malaysia Deepwater Industry Collaboration towards Prudent Surveillance Practices and Excellence Production Sustainability

Malaysia's deepwater fields among the significant contributors of Malaysia's total oil production. Success of deepwater projects needs integrated production surveillance plan to provide holistic approach for production system evaluation and optimization over a wide range of operating conditions from sub-surface to the export point. Following to reservoir depletion phase, it is essential to implement integrated production surveillance

plan in providing holistic approach for production system evaluation and optimization from sub-surface to the export point while managing deepwater assets. As the host authority overseeing Malaysia's hydrocarbon resources, PETRONAS Malaysia Petroleum Management (MPM) has established a multidisciplinary strategic initiative known as Deepwater Industry Collaboration (DWICON) to address key challenges across Exploration, Development, and Production Operations as outlined in Figure 1. Recognizing that deepwater assets contribute approximately one-third of the national production, DWICON adopts a holistic approach to safeguard this critical resource by fostering industry-wide collaboration, enhancing operational excellence and championing sustainable development across Malaysia's deepwater sector. Under DWICON, a focused taskforce called Deepwater Integrated Surveillance Excellence (DWISE) is established not only to assure national production target delivery for all deepwater asset in Malaysia but also to solidify existing industry collaboration towards prudent surveillance and production sustainability. DWISE approaches started with the consolidation of common challenges specific to Malaysia deepwater production issues and solutions in six (6) focus workstreams as showed in Figure 2 namely sand risk management, flow assurance, artificial lift, subsea well intervention, chemical management and subsea inspection and maintenance.

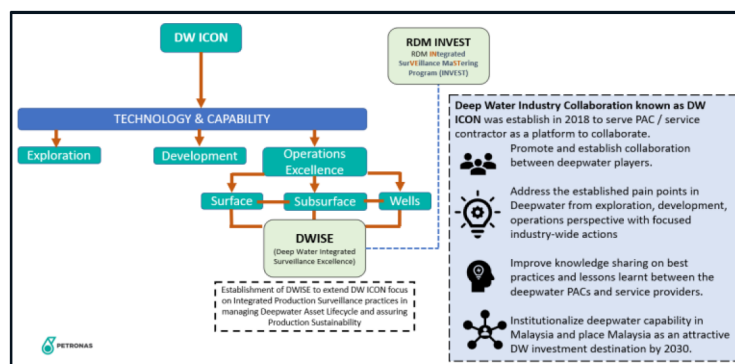


Figure 1—Deepwater Industry Collaboration (DW ICON) pillars and workstreams.

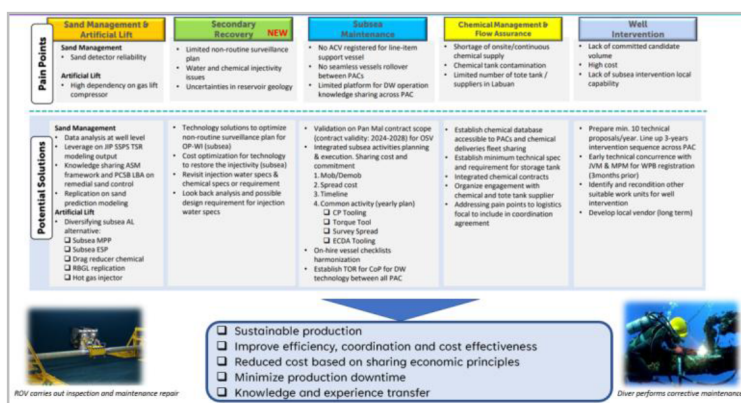


Figure 2—DWISE Team Members Comprise of Multidisciplinary Workstreams

Driving Deepwater Asset Lifecycle Sustainability Through DWISE Integration

Figure 3 explains that DWISE team comprises not only key stakeholders such as MPM and deepwater operators but also includes service providers relevant to each workstream, bringing extensive experience from both shallow and deepwater operations globally. This diverse collaboration enables the integration of innovative solutions and cutting-edge technologies, driving continuous improvement and enhancing the

efficiency, safety, and sustainability of deepwater asset management. DWISE deliverables is divided into 2 phases:

1. Phase 1: Development of Deepwater Surveillance Playbook. Consolidation of common deep water assets production challenges and solutions in four focus areas i.e., sand risk management, flow assurance, artificial lift, subsea well intervention, chemical management and subsea inspection and maintenance.
2. Phase 2: Enablement of Integrated Common Services in Subsea activities. i.e., Well Intervention, Chemical Management and subsea inspection and maintenance.



Figure 3—Seamless Integration between Industry Players is Critical in Managing Deepwater Asset Lifecycle Sustainability

The brief of DWISE Phase 1 and Phase 2 deliverables is demonstrated in Figure 4.

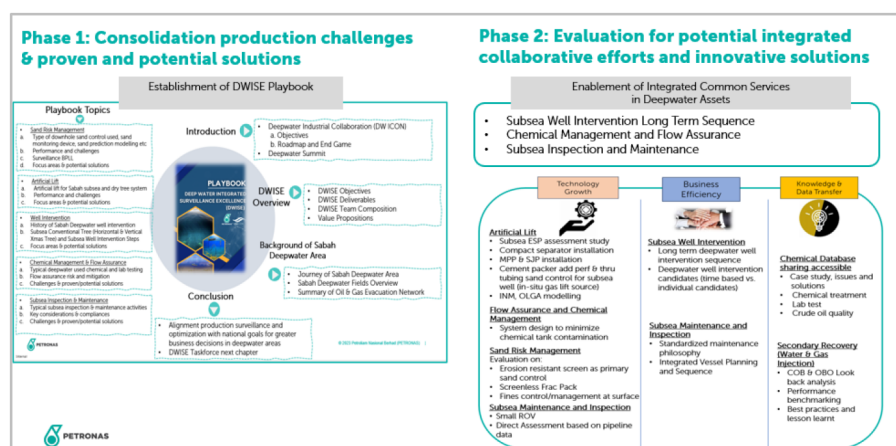


Figure 4—DWISE Phase 1 and Phase 2 Deliverables

The establishment of DWISE has the potential to create significant value for Malaysia deepwater oil and gas industry through its core objectives:

1. Knowledge Transfer on Deepwater Surveillance Practices: By facilitating knowledge transfer among MPM, operators and service providers it can lead to proactive problem-solving and production optimization subsequently minimize the down time in production.
2. Enrich Proactive and Seamless Production Conversation: The enriched communication between MPM and operators can drive towards effective collaboration and information sharing, DWISE

can help streamline production-related discussions and decision-making processes. This can lead to improved coordination and efficiency in deepwater operations.

3. Identification and Pursuit of Opportunities and Threats Mitigation: The proactive and collective approaches can create value by improving operational efficiency, increasing safety, optimizing production processes, reducing costs, and enhancing risk management.
4. Long Term Integrated Common Services for Deepwater Asset Planning: With integrated contracting solutions, subsea vessel sharing and utilization as well as common inspections and maintenance, it can further improve deepwater asset economic viability and production uptime.

Background on Malaysia Deepwater Fields and Facilities System

The deepwater journey of Malaysia's oil and gas industry has been marked by significant advancements in exploration and production, particularly in the region of Sabah. This sector has seen remarkable growth since the initial discovery in the late 1990s. The exploration efforts in Sabah's deepwater fields have yielded remarkable results, with several significant discoveries being made since late 1990s. Notably, substantial discoveries have been situated within the Upper Miocene Cycle V reservoirs of two primary areas: the Baram Delta and the Northwest Sabah Province.

In summary to the deepwater facilities and gas and oil evacuation point for each field as illustrated by Figure 5, all fields have manifold subsea facilities except for field B, which also the only field that uses dry Christmas Tree. However, Field C uses both dry and wet Christmas Tree while all other fields are using wet type. The final processing hub for Field A is at semi sub–Floating Production System (FPS) while Field B at fixed platform. Both fields deliver oil to the same location, Onshore Oil and Gas Terminal, but only gas in Field A is being re-injected while gas in Field B is being delivered to Onshore Gas Terminal alongside with gas from Field C and D. The final processing hub field C and D is at FPSO which also the Delivery point for oil from both fields. On the other hand, oil and gas from Field E are being processed and delivered to FLNG.

Operator	Field	Christmas Tree Type	Subsea	Final Processing Hub	Oil Delivery Point	Gas delivery Point
Operator X	Field A	Wet	Manifold	Semi Sub FPS	Onshore Oil and Gas Terminal	Re-injected
Operator X	Field B	Dry	N/A	Fixed Platform	Onshore Oil and Gas Terminal	Onshore Gas Terminal
Operator Y	Field C	Dry and Wet	Manifold	FPSO	FPSO	Onshore Gas Terminal
Operator Y	Field D	Wet	Manifold	FPSO	FPSO	Onshore Gas Terminal
Operator Y	Field E	Wet	Manifold	FLNG	FLNG	FLNG

Figure 5—Summary of Deepwater Facilities in Sabah

Artificial Lift Methods Implemented in Malaysia's Deepwater Fields

In Malaysia's deepwater operations, maximizing hydrocarbon recovery often requires the use of artificial lift, particularly in depleted assets where wells are unable to flow naturally to sustain optimal production. Both wet tree subsea wells and Dry Tree Unit (DTU) wells present unique challenges in artificial lift deployment, driven by differences in well completion design, well accessibility, intervention requirements, and operating environments. Wet Tree wells, typically completed with subsea X-Mas trees on the seabed, often demand remote or subsea-compatible lift solutions that can be deployed via riser less or riser-based intervention systems. In contrast, dry tree wells usually located on Tension Leg Platforms (TLPs) or Spars

which allow more direct vertical access, enabling a broader range of artificial lift options and intervention methods. Over the years, Malaysia's deepwater operators have evaluated & implemented a variety of artificial lift methods from gas lift and electric submersible pumps (ESPs) to more specialized techniques such as Riser-Based Gas Lift (RGL), Multiphase Pump (MPP) with Compact Separator, Subsurface Gas Lift (SSGL), Coiled Tubing Gas Lift (CTGL) and Coiled Tubing ESPs (CTESP). These solutions are tailored to address declining reservoir pressure, increasing water cut, and complex flow assurance challenges, ensuring production continuity and economic viability in one of the most technically demanding offshore environments.

Deepwater fields typically have commingled flow of several wells in a single riser, increased internal/external hydrostatic pressure and low ambient temperatures. The increased hydrostatic pressure from the liquid column in the riser results in higher back pressure on the wells, which can lower the recoverable reserves. Deepwater fields often require an artificial lift method to either add mechanical energy or lighten the fluid in the riser at the riser base. Subsea boosting and electric submersible pumps provide mechanical means for adding energy in the system. Although these two artificial lift methods provide high-rate capability, they consist of several system parts which can be damaged by solids in the fluids and require a highly reliable electric power system. Gas lift systems, comparatively, are less complex and are typically the artificial lift method of choice when gas is available at a suitable pressure (*OTC-31108-MS*).

Background Field C and Field D

Field C and Field D are 2 deepwater oil fields with tie-back to the same FPSO. Field C is located 120km north-west of the island of Labuan, off the coast of Sabah, offshore East Malaysia, in water depths of approximately 1,330m. The oil and gas field, the first deepwater development in Malaysia, was developed with floating production storage and offloading (FPSO) vessel. Field C as showed in [Figure 6](#) is a mixture between dry tree wells and subsea tree wells comprise of oil producers, gas injector and water injectors. Dry tree wells onboard Spar produce to FPSO via 3 floating production flowlines to FPSO. Subsea wells produce via 2 production flowlines to FPSO. Field D is all subsea oil producer and subsea water injector wells with long tie-back of 2 production flowlines to FPSO. The produced gas is exported to onshore Gas Terminal via gas export pipeline while excess gas is disposed into subsea gas injector well in Field C.

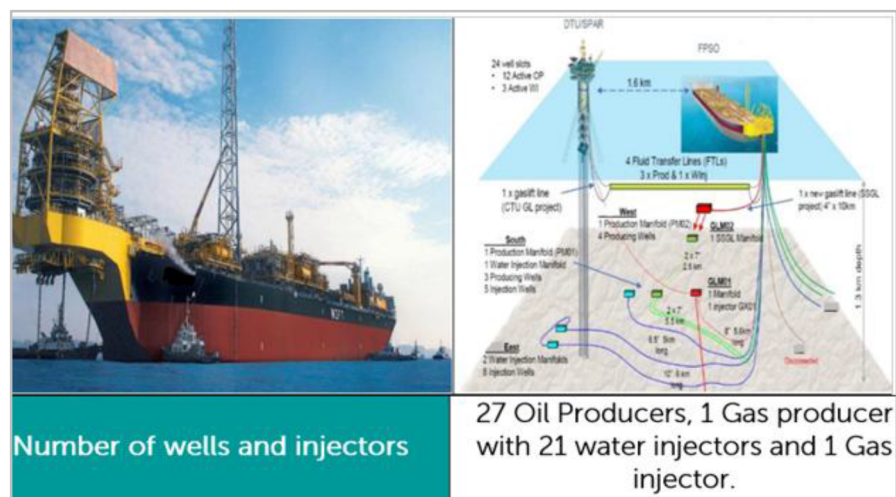


Figure 6—Field C Info & Facilities Layout

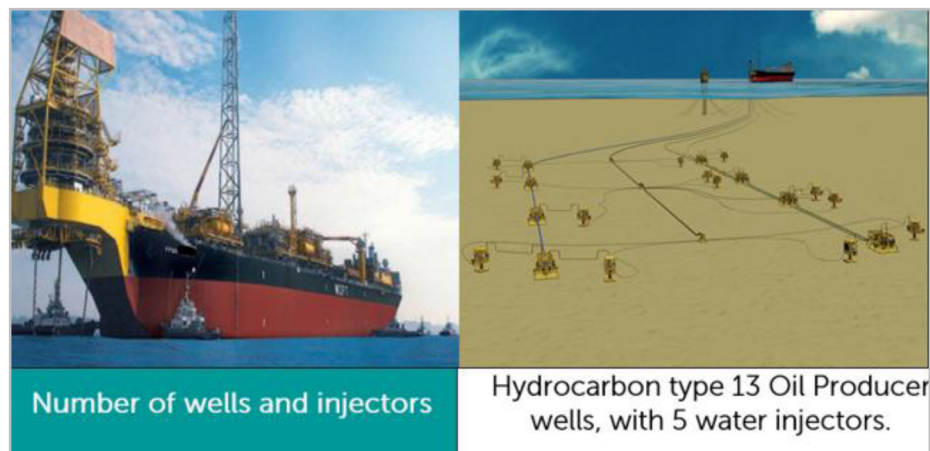


Figure 7—Field D Info & Facilities Layout

The wells struggled to sustain production due to reservoir pressure depletion and increased in watercut. It became worse with high backpressure due to severe flowline slugging in combination and difficulties of kick off the well after shut-in. Initiative to improve well recovery was explored focusing on improving well lifting, riser lifting and lowering well backpressure.

Artificial Lift for Dry Tree Wells & Topside Pressure Boosting System

Field C has experienced and implemented few methods to kick off and to lift the production for Dry Tree Unit (DTU) wells. Here are the methods:

1. The first method explored was using temporary well test package (Refer [Appendix 1](#)) to lower the backpressure to kick off well. The well later switched back to flow transfer line after it stabilized. This practice was discontinued
2. Coil Tubing Electric Submersible Pump (CTESP)(Refer [Appendix 3](#)) pilot project was implemented in one of the dry tree well with production gain around 650 BOPD in October 2017 but the unit unable to start due to downhole failure in August 2018. The CTESP string was retrieved and the well then converted into CTGL.
3. A pilot test with small Surface Jet pump (SJP) (Refer [Appendix 4](#)) installed in Spar Test header as part of evaluation to reduce dry tree wells backpressure using water injection as motive fluid. After achieving good results, 2 bulk SJPs are installed in HP1 and IP production headers and still online until today. With current well production line-up, around 200 psig well backpressure reduction is observed thus benefitting low pressure wells which contributed to higher production, shorter time requires for well kick off and more stable multiphase flowlines to FPSO as motive fluid.
4. Another method explored was using Coil Tubing gas lift (CTGL) (Refer [Appendix 5](#)) for dry tree wells. This method is still in use until today. Now, out of 11 CTGL wells, only 1 well can still flow naturally in the event of supply outage.
5. One option as temporary measure is to inject nitrogen gas via chemical injection line into chemical Injection mandrel downhole to kick-off weak well. This method provides the deepest injection point in the well (Refer [Appendix 2](#)) but requires equipment mobilization from shore and still being practise until today.
6. The latest pressure boosting project is Compact Separator system. It comprises of two phase separator with multiphase pumps (Refer [Appendix 6](#)) to FPSO. Compact Separator system can operate in series at downstream of SJP for additional boosting. About 600 BOPD initial gain observed from this new facility.

Artificial Lift for Subsea System

During Field Development, there was no consideration of installing artificial lift for subsea production system of Field C. However, for Field D which was put online later than Field C, already incorporated artificial lift system in Field Development. None of the subsea wells from Field C and D are equipped with downhole artificial lift system.

Subsea Gas Lift (SSGL)

In Field C, SSGL (Refer [Appendix 7](#)) system was implemented to inject gas lift into subsea flowline. SSGL Phase 1 experienced hydrate block after about 3 months online and was replaced with the improved design of SSGL Phase 2 in Dec 2024 which is still online until today. About initial 650 BOPD oil gain was achieved contributed by reduction in backpressure, more stable flowline and reactivation of low pressure well. SSGL also eliminates the requirement to operate the wells in cyclic mode i.e. wells online in 20 days followed by shut-in by 10 days as was practiced earlier. Further fine tuning in production optimization including activation of second subsea flowline is on the way.

Riser Base Gas Lift (RBGL)

Based on the lessons learned from the implementation of the SSGL system in Field C, the design approach for Field D's subsea production system was modified. Field D is using different approach for supplying high pressure gas lift into subsea production system as it was already considered at design stage. High pressure gas from the FPSO is supplied to the gas lift injection points at Riser Base Gas lift manifold via umbilical. As the reservoir depletes and increasing water cut, reliance on RBGL becoming more important.

Operational Challenges

Well Test Package

The well test package is designed as a temporary arrangement to facilitate the kick-off weaker wells. There's a safety concern related to the direction of the wind affecting the operation of a single flare. The wind direction can impact how safely and effectively the flare operates and the flaring activity releases gases into the atmosphere, which can have negative environmental impacts. Moreover, the available surge tank and the capacity of the multi-phase pump has limited the application of well test package especially when dealing with varying fluid flow rates and pressures from the wells. The layout of the equipment for well testing package requires significant space in DTU area. This could have implications for the available workspace and logistics on the DTU footprint. Renting or leasing the necessary equipment for well testing can be costly. This suggests that budget considerations are a factor in planning and executing the well test package.

Nitrogen Gas Injection via Downhole Chemical Injection Mandrel

Temporary injecting nitrogen gas downhole through existing chemical injection mandrel can expedite kick-starting of the weak wells that are difficult to unload due to low reservoir pressure. The downside of this approach requires proper logistics for transporting personnel, nitrogen gas and the associated equipment package from onshore to the well site and any delays in the process could lead to delays in bringing the well online. The injection of nitrogen gas via small internal diameter of the 3/8" chemical injection line deep into the well causing high pressure loss and limits the maximum injection rate below 1 MMscfd on N₂. To expedite the weak well kick off process further, a dual-lift approach is sometimes employed. This involves simultaneously injecting nitrogen gas using the chemical injection mandrel and using gas lift via CTGL.

CTESP

As for pilot test CTESP string was installed with BHA depth is above SCSSV. By the time CTESP string was retrieved, implementation of CTGL project already underway and priority to convert the wells into CTGL is higher rather than to continue with CTESP.

SJP

The operation of surface jet pump depends on the availability of water injection pumps on the FPSO. Total water injection consumption for running SJPs can be as high as 50,000 BWPD hence it may occupy total water injection availability for water injection wells if the pumps are not at full capacity. During planned or forced shut-in, the production line is filled with injected water. Therefore, it is difficult to start up the well when the line is filled with seawater, especially if the required water injection pump or the gas lift is not yet operational. The operation of the surface jet pump also introduces additional water into the production system which additional load to produced water treatment system at FPSO. SJP nozzle or 'Insert' can be changed to meet new requirement i.e. to operate in series with Compact Separator downstream of SJP.

CTGL

CT L is used to assist in unloading and maintaining production from wells with low Productivity Index (PI) and/or low pressure against the topside pressure. The length of CTU GL is limited to around 50 meter above the depth of the Subsurface Controlled Safety Valve (SCSSV), which can make the lifting process less efficient. When future well interventions are required, the CTU GL string must be pulled out from the well before any well entry activities can take place. This process of pulling out of the hole (POOH) and re-running in hole (RIH) takes additional time. Every time when CTU string is pull out and reinstall, approximate 40–50-meter section of CTU string will be cut to allow make-up of CTU hanger thus, the stings become shorter over time.

The gas used for gas lift, which is supplied from the gas export line, has a low temperature (around 3-4 degrees Celsius). Continuous methanol injection at the incoming gaslift riser at DTU is currently in practice as to reduce the hydrate formation. This low temperature can cool down the flowing wellhead temperature. In some cases, this cooling effect can cause the flowing wellhead temperature from certain wells to become cold, leading to the need for continuous chemical injection of Pour Point Depressant (PPD) chemical injection at the Christmas tree. It's necessary to respond quickly to prevent recurring issues of hydrate or wax blockages in the production tubing due to the cold temperature of the gas lift used in CTU GL. This can involve securing the well via bullheading a tubing volume of diesel until safety valve depth to prevent these blockages. A potential issue arises where production fluids ingress into the Coiled Tubing Unit (CTU) due to the passage of a check valve in the CTU Bottom Hole Assembly (BHA). This ingress of production fluids can create flow assurance issues, such as hydrate plugging within the CTU GL system.

Compact Separator and Multiphase Pump

This project is intended to provide additional reduction in well backpressure downstream of the existing Surface Jet pumps. Dry tree low pressure wells will have options either to flow either via SJP1 or SJP3 before entering the common inlet line into 2 phase Separator. Liquid from separator liquid outlet will be pump via multiphase pump to FPSO via FTL01 as single phase while the gas separator gas outlet will flow to FPSO into IP separator via FTL03 as single phase. Single phase flow in FTL1 and FTL3 will eliminate riser slugging issue. Amount of liquid from well streams plus from SJP power fluids will be limited by Multiphase pump flow capacity of 75,000 BLPD hence will requires optimization. High pressure wells will have option to flow directly to FPSO via dedicated HP2 multiphase flowline

SSGL

There was no consideration to install gas lift injection system at subsea production in the original design of Field C subsea production system. SSGL phase 1 system was added using unused well production jumper hub in Production Manifold 2. Methanol injection dosage into SSGL phase 1 was relying on the injection point at turret. Due to hydrate blockage the injection line is no longer in used and to be replaced by SSGL phase 2. SSGL phase 2 is comprise of 1 new 4" × 10km gas lift line supply line from FPSO, 2 SSGL injection lines at jumper PX08 and PX06, subsea pressure control valves, dedicated subsea methanol injection systems and heat exchangers to mitigate hydrate formation. The project is currently on-going.

RBGL

Riser Base Gas Lift system is part of subsea production system for Field D. Instead of using gas lift pipeline, gas lift is supplied from FPSO via 2 times small umbilical cores 4 × 0.75" ID which is insulated. A dedicate methanol injection points design also included in the design of riser base gas lift system for each Flowline 1 PP01 and Flowline 2 PP02. There is a risk of hydrate blockage in the umbilical which has occurs in the past. Currently RBGL for PP01 is isolated due to gas leak and still pending for replacement.

Table 1—Methods deepwater fields have implemented in Malaysia.

Field	AL Method	Status	Challenges	Best Practices / Lesson learnt
Field A	No AL	N/A	Managing GOR	Manage high GOR through RMP Optimization
Field B	GL	Good	None.	None.
Field C	No AL	N/A	Backpressure impacting weak well to kick off.	As temporary measures, well test package to reduce backpressure during kick off.
Field C	SJP	Good	More water handling in production system & reliance on water injection pump availability.	Flexible operating envelopes by changing SJP Insert and able to operate in series with Compact Separator to increase pressure boosting.
Field C	CTESP	Stop Working	Downhole power failure.	This method proven but Field C decided to switch to CTUGL since the gas lift supply available during that time.
Field C	CTGL	Good	Shallow injection depth & impose additional challenge for any well intervention	To utilize deep chemical injection line to kick-off weak well using nitrogen gas. Consideration to convert to GL mandrel is not feasible with current Dry Tree well design.
Field C	SSGL Phase 1	Stop working	Hydrate block suspected at subsea area.	To use dedicate system to supply gas lift into subsea system equipped with sufficient monitoring and hydrate prevention/mitigation system.
Field C	SSGL Phase 2	Good	Hydrates.	SSGL Phase 2 equipped with subsea heat exchanger, pressure & temperature sensors & methanol injection and dedicated operating procedures. Having 2 SSGL streams provide operational redundancy which can allow gaslift supply into dual flowline mode or single flowline mode.
Field C	Compact Separator	Good	1. Depending on Power Supply from FPSO. 2. Cannot tolerate sand 3. Limited by pump fluid capacity. 4. High backpressure in gas line from Compact separator outlet to FPSO during Compact Separator start-up due to presence of liquid in the line.	1. Current power uptime improved with sufficient back up. 2. Manage SJP to reduce water. 3. Wells with high sand production are bypassing Compact Separator. 4. Standard procedure to flush liquid in the gas line using gas lift supply from shut-in well prior to start-up of Compact separator to dewater the line. *DTU compact separator will improve slugging impact and reduce back pressure to well.
Field D	RBGL 1	Good	1. RBGL leak. 2. RBGL hydrate blockage.	Periodical methanol dosage into RBGL umbilical to clear the line.

Consideration for artificial lift system for deepwater field are different and unique depending on reservoir properties, water depth, field location & etc. Sabah deepwater field has shown the different requirements and the needs for artificial lift.

Potential Best Practices to Enhance Artificial Lift Deployment & Kick off Solutions for Wet & Dry Tree Well

To sustain and enhance production performance in both wet tree (subsea) and dry tree (topside) wells, the following potential best practices have been identified as opportunities to optimize artificial lift deployment and kick-off solutions, supporting both current operations and future project execution:

- Adopt standardized evaluation templates for artificial lift feasibility, potentially enabling more consistent technical and economic screening across different deepwater well types and operating conditions. The template should factor in deepwater-specific constraints such as subsea tie-back distance, riser configurations, wellbore geometry, reservoir fluid properties, and host facility capacity, ensuring more informed and comparable decision-making.
- Establish a centralized performance database of historical artificial lift/kick-off jobs, which could serve as a reference for future lookbacks, benchmarking, and accelerated decision-making for both new and mature wells. By indexing cases according to water depth, completion type, reservoir characteristics, and intervention method, the database would facilitate robust lookbacks, enhance benchmarking accuracy, and speed up the decision-making process for both brownfield reactivations and greenfield start-ups.
- Incorporate multi-solution readiness into offshore facility design so that topside and subsea systems can, where feasible, be adapted to switch between lifting methods with minimal downtime or modification. This includes provisions for spare gas lift mandrels, umbilical/hydraulic line routing, ESP power supply tie-ins, and subsea control system compatibility.
- Strengthen flow assurance monitoring capabilities through potential use of predictive diagnostics, which may help in early detection and mitigation of hydrate, wax, or scale formation risks in both wet and dry tree production systems. Coupling predictive diagnostics with automated control logic can trigger timely remedial actions such as chemical injection, pigging, or flow rate adjustment. In Malaysia's deepwater environment, where long tiebacks and complex seabed profiles heighten hydrate and wax risks, early detection is critical for maintaining lift efficiency and preventing unplanned shutdowns.
- Facilitate structured cross-asset knowledge sharing between operators, service providers and PETRONAS to capture lessons learned, with the aim of accelerating replication of successful lift strategies and troubleshooting methods.

Conclusion

Artificial lift plays a critical role in sustaining Malaysia's deepwater production, where challenging reservoir conditions and high operating costs demand precise technology selection, operational readiness, and integrated planning. Experience across Malaysia's wet and dry tree wells shows that success hinges on technical preparedness, operational agility, and early adoption of lessons learned.

Wet tree wells require robust subsea adaptability and reliable remote operability, while dry tree wells benefit from surface accessibility but still demand proactive lift strategies for low-pressure or low Productivity Index (PI) conditions. Embedding best practices into project planning, facility design, and surveillance not only reduces intervention cycle time but also maximizes recovery, defers abandonment and enhances economic returns aligning with PETRONAS long-term production sustainability goals.

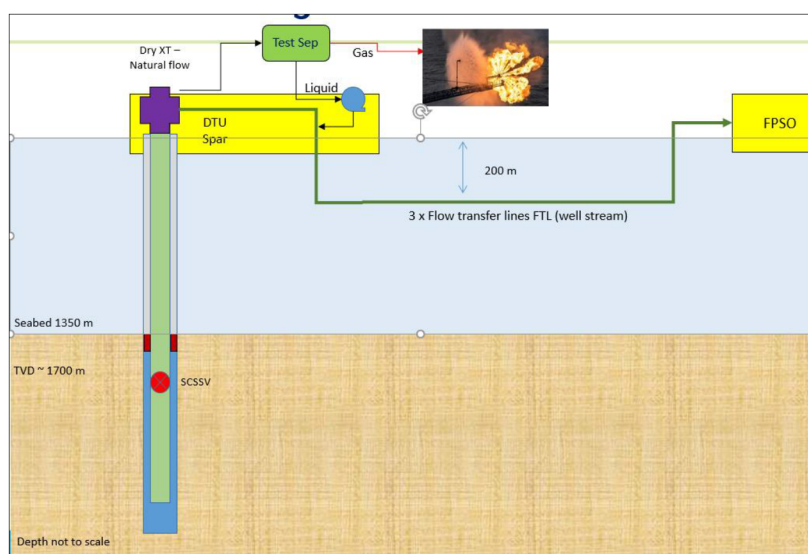
Achieving this business goal requires close collaboration between operators, service providers, and regulators, supported by structured surveillance programs that integrate risk assessment, uncertainty management, and opportunity capture from reservoir to facility. This unified approach will strengthen Malaysia's production resilience, safeguard asset integrity, and create enduring value across the deepwater asset lifecycle.

References

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2. PETRONAS Deepwater Integrated Surveillance Excellence (DWISE) Playbook 2024

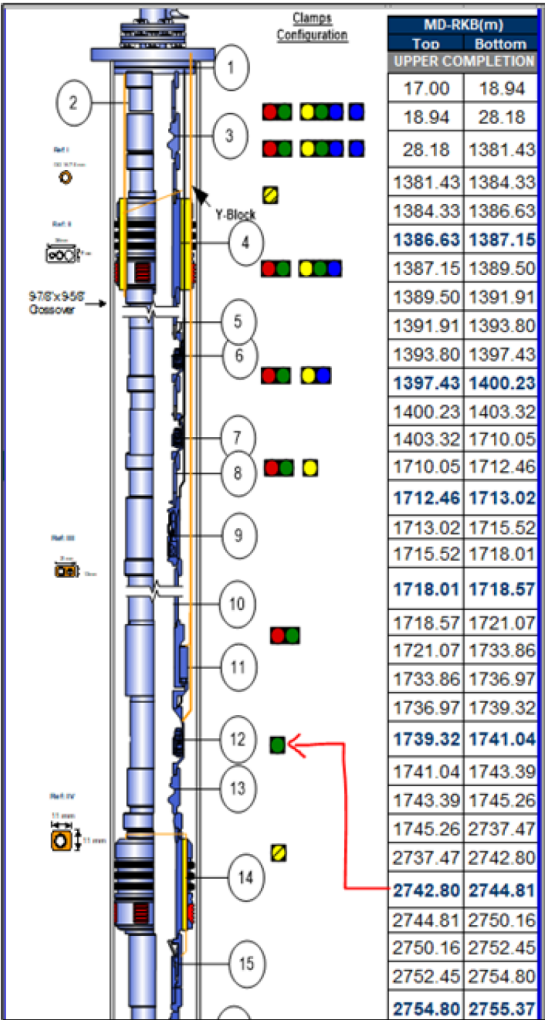
Appendix 1

Temporary well test package used to kick-off weak well



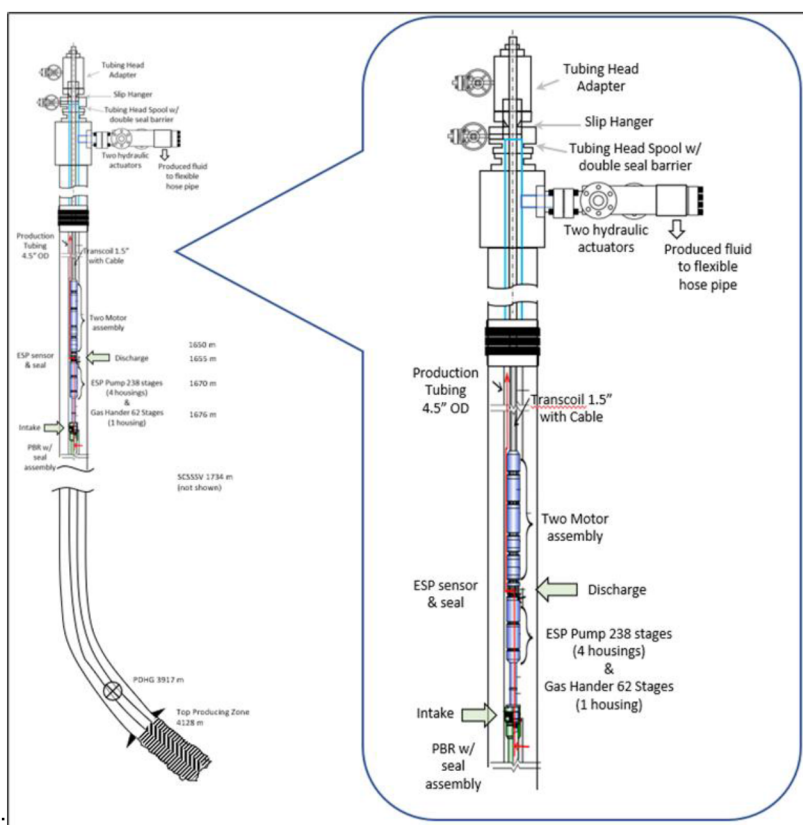
Appendix 2

Nitrogen gas lifting via deep chemical (Scale inhibitor) mandrel to assist in kick-off weak Dry Tree wells in



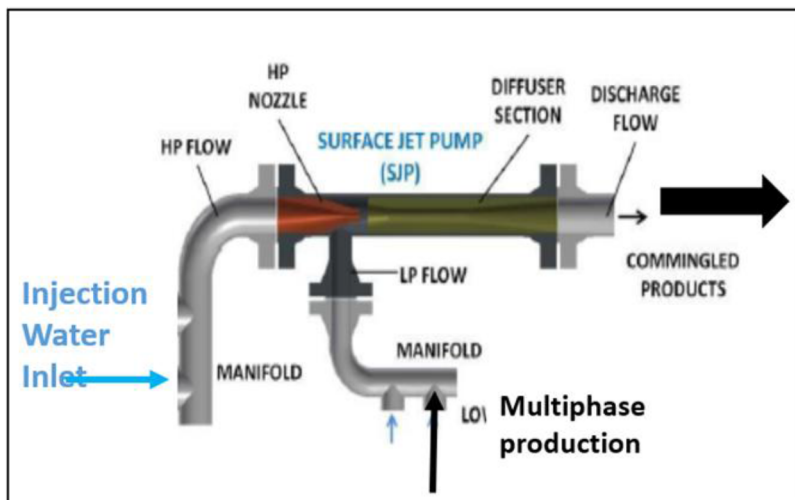
Appendix 3

Dry Tree Unit (DTU) well Electric Submersible Pump (ESP) in Field C



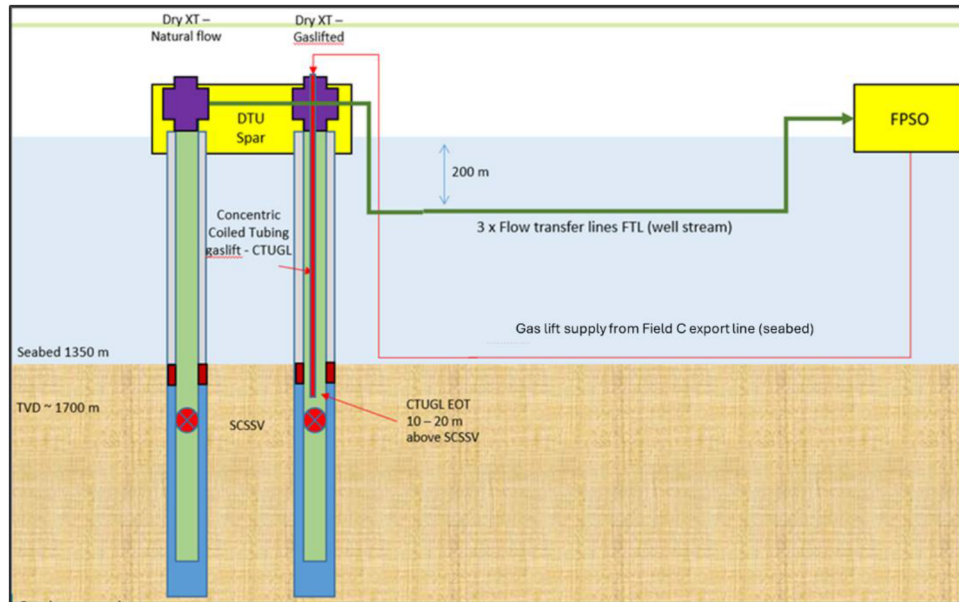
Appendix 4

2 x Surface Jet Pumps for HP1 and LP manifolds and 1 x smaller SJP in test manifold onboard Spar in Field C



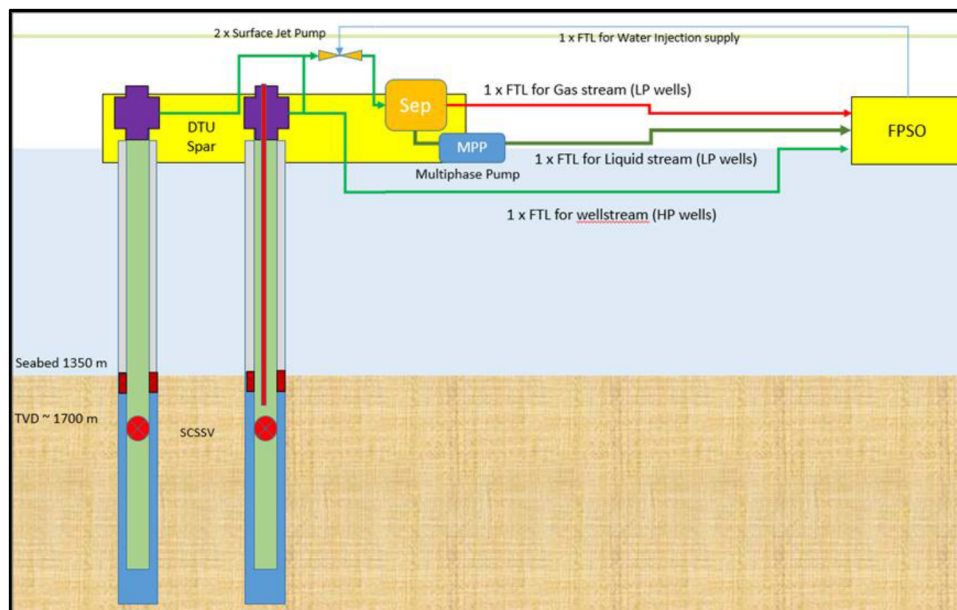
Appendix 5

Dry Tree Coiled Tubing Gas Lift as permanent setup for weak well on the Spar in Field C



Appendix 6

Compact Separator with Multiphase Pump to compliment SJP and reduce backpressure to the well in Field C



Appendix 7

Subsea Gas Lift (SSGL) Phase 1 line tie-in at empty well slot at Field C. The connection is now removed for tie-in of new infill well

